

PONDO Risk Profiling – ID-Based Logic Specification (Age & Gender Extraction for Risk Scoring)

1. Overview

This document defines the logic required to extract Age and Gender from a South African ID number and convert these into risk scoring inputs for the PONDO platform. This module forms part of the risk profiling engine and should be implemented as a reusable function or service.

2. South African ID Structure

A valid South African ID number consists of 13 digits: YYMMDD SSSS C A Z

Relevant fields:

- YYMMDD (Digits 1–6): Date of Birth
- SSSS (Digits 7–10): Gender indicator

3. Core Logic Flow

- Step 1: Validate ID format (13 digits, numeric only)
- Step 2: Extract DOB (yy, mm, dd)
- Step 3: Determine century (1900/2000 logic based on current year)
- Step 4: Validate date
- Step 5: Calculate age
- Step 6: Extract gender (≥ 5000 Male, else Female)

4. Output Structure

```
{  
  "valid": true,  
  "age": 56,  
  "gender": "Male"  
}
```

5. Risk Scoring Conversion

Age → Risk Score:

<18: Reject
18–25: 80
26–40: 50
41–60: 20
60+: 10

Gender → Risk Score:

Male: 10
Female: 0

6. Python Reference Implementation

```
from datetime import datetime

def extract_risk_inputs(id_number: str):
    if not id_number.isdigit() or len(id_number) != 13:
        return {"valid": False}
    try:
        yy = int(id_number[0:2])
        mm = int(id_number[2:4])
        dd = int(id_number[4:6])
        current_year = datetime.now().year % 100
        year = 1900 + yy if yy > current_year else 2000 + yy
        dob = datetime(year, mm, dd)
        today = datetime.today()
        age = today.year - dob.year - ((today.month, today.day) < (dob.month, dob.day))
        gender_digits = int(id_number[6:10])
        gender = "Male" if gender_digits >= 5000 else "Female"
        if age < 18:
            return {"valid": False, "reason": "Underage"}
        elif age <= 25:
            age_score = 80
        elif age <= 40:
            age_score = 50
        elif age <= 60:
            age_score = 20
        else:
            age_score = 10
        gender_score = 10 if gender == "Male" else 0
        return {
            "valid": True,
            "age": age,
            "gender": gender,
            "age_score": age_score,
            "gender_score": gender_score
        }
    except:
        return {"valid": False}
```

7. Integration Example

```
profile = extract_risk_inputs(id_number)

if not profile["valid"]:
    reject_transaction()

risk_score += profile["age_score"]
risk_score += profile["gender_score"]
```

8. Key Design Principles

Age is a primary risk signal. Gender is a secondary (low-weight) signal. The module should execute in real-time (<50ms).

9. Security & Compliance

Do not store raw ID numbers in plaintext. Encrypt sensitive data (AES-256 recommended). Mask ID numbers in logs. Ensure POPIA compliance.

10. Edge Case Handling

Invalid date: Reject
Underage: Reject

Non-numeric ID: Reject
Incorrect length: Reject

11. Final Note

This module provides a fast, zero-cost identity risk signal to enhance decision-making at checkout.